

Computer Vision Practical Report:

EDGE DETECTION USING CANNY METHOD

1. AIM

To read an input image using OpenCV, detect its structural edges using the Canny edge detection algorithm (cv2.Canny()), display the original and edge-detected images side-by-side using Matplotlib, and save the result to disk.

2. PROCEDURE

- Load the input image and convert it to grayscale using cv2.cvtColor().
- Apply Gaussian blur using cv2.GaussianBlur() to reduce image noise.
- Apply Canny edge detection using cv2.Canny() with selected lower and upper threshold values.
- Display the original and edge-detected images side-by-side using Matplotlib.
- Save the edge-detected output image to disk using cv2.imwrite().

3. PROGRAM CODE

```
import cv2

image = cv2.imread(r"C:\Users\adminuser\Desktop\Python CV\E16-IMG.png")

if image is None:
    print("Image not found")
    exit()

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

blur = cv2.GaussianBlur(gray, (5, 5), 0)

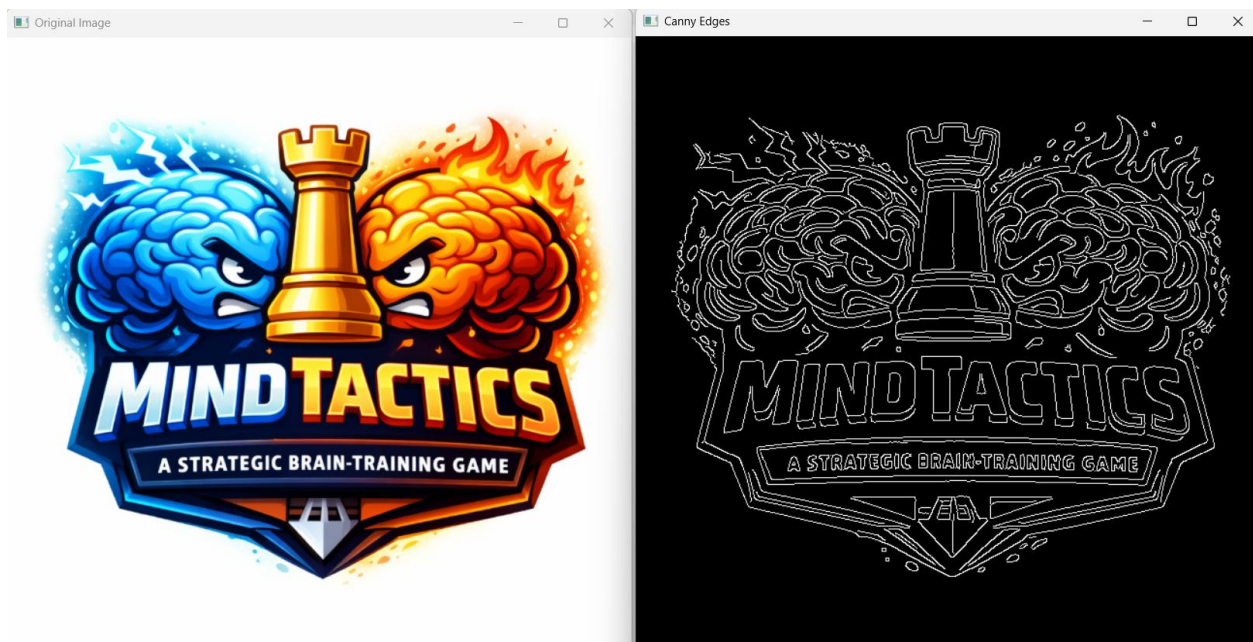
edges = cv2.Canny(blur, 50, 150)
```

```
cv2.imshow("Original Image", image)
cv2.imshow("Canny Edges", edges)

cv2.imwrite("canny_edges.jpg", edges)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.